



LAPI GANESHA UTAMA • LAB IOT & FISIKA ITB • PERTAMINA

Datasheet Sistem GLD

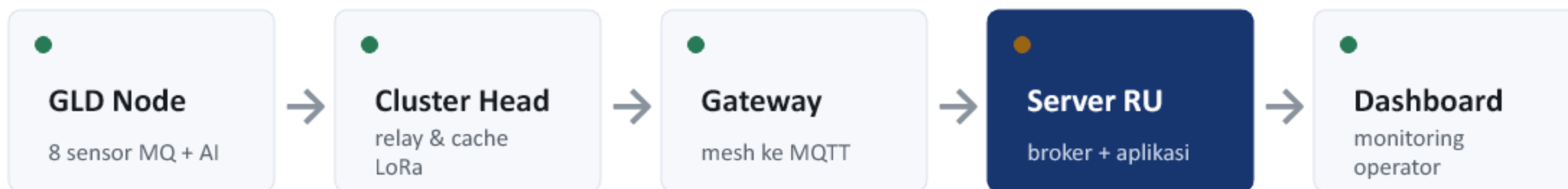
Spesifikasi arsitektur, interface, data flow, aplikasi, dan infrastruktur Kilang

REV 0.8 • 19 AGUSTUS 2026 • DRAFT TECHNICAL SPECIFICATION

Tangki LPG & SRU | GLD Tahap 2

Arsitektur operasional terdiri dari lima lapisan

Inferensi dijalankan pada GLD Node; telemetry ditransmisikan melalui LoRa dan MQTT ke aplikasi RU.



KILANG

Arsitektur, ATEX, cakupan gas

LAPANGAN

Perangkat, mounting, catu daya

IT PERTAMINA

Server, MQTT, keamanan jaringan

System status: RU NOT COMMISSIONED

ENGINEERING TEST: PASS. Hasil uji laboratorium bukan bukti operational acceptance di RU.

- **Firmware integration test**
PASS — 8-CH mesh dan automatic alarm push diuji di kampus
- **GLD power interface**
BASELINE — 24 VDC ≥ 2 A per unit; cable dan PSU oleh RU
- **Mechanical mounting design**
REFERENCE — L-bracket/U-clamp; no drilling; no welding
- **AI execution status**
REPORTED ON-DEVICE — 99,20%; firmware evidence belum tersedia
- **RU deployment package**
AVAILABLE — broker, backend, frontend, installer; not commissioned
- **Network integration package**
AVAILABLE — MQTT contract, security baseline, go-live checklist

STATUS SISTEM

NOT COMMISSIONED — operasi RU memerlukan server, network path, site permit, dan sertifikasi hazardous-area.

Tiga prerequisite wajib ditutup sebelum instalasi

Setiap prerequisite memerlukan owner, technical approval, dan evidence of closure dari Kilang.

1 Server & jaringan

- Sediakan server/VM RU
- Registrasi MAC Gateway
- Satukan subnet GW + server
- Tetapkan VLAN/firewall/802.1X

2 Instalasi & daya

- Sediakan kabel + PSU 24VDC
- Konfirmasi struktur mounting
- Setujui bracket dan fastener
- Verifikasi solar + baterai CH

3 Izin & sertifikasi

- Sahkan TRA/JSA
- Sertifikasi ATEX
- Izin ketinggian permanen
- Siapkan sampel gas uji

GAS-DETECTION REQUIREMENT

Tetapkan mapping Benzena, CO, dan H₂S terhadap output class model sebelum test protocol dibekukan.

Kilang: server, network, dan operational controls

Baseline sizing berlaku untuk pilot; final sizing mengikuti device count, data rate, retention, dan availability target.

SERVER / VM RU

- **Minimum**
4 vCPU • 8 GB RAM • 100 GB SSD
- **Rekomendasi**
8 vCPU • 16 GB RAM • 250 GB SSD
- **Platform**
Ubuntu Server 22.04 LTS • Docker Compose
- **Antarmuka**
1 GbE • IP stabil / reservasi DHCP • DNS/NTP

JARINGAN & OPERASI

- **Akses**
Gateway dan server dapat saling menjangkau pada VLAN yang disetujui
- **Keamanan**
Registrasi MAC, firewall, ACL MQTT, TLS/VPN, akun servis
- **Operasi**
PIC IT/OT, backup, retensi, monitoring, patching dan incident path
- **Keputusan**
Host/port MQTT produksi; endpoint operator; izin koneksi VPS/HQ

PRODUCTION PARAMETERS —

OPEN

Port 1884 hanya untuk bench. IT/OT Kilang wajib menetapkan port, certificate chain, DNS/IP, backup, dan retention produksi.

DFD-01 — akuisisi sensor hingga dashboard RU

Processing dan visualization beroperasi lokal di RU; koneksi VPS tidak menjadi dependency operasi lokal.



NOT IMPLEMENTED

Adapter MQTT → apps/backend tidak terdapat pada repo aplikasi. Implementasi dan integration test bersifat mandatory.

Batas interface: MQTT/Node-RED → backend GraphQL

PertaminaGLD menghasilkan normalized MQTT topics; aplikasi menerima data melalui ingestion adapter.

PertaminaGLD

- **Input**
gld/gateway/uplink • raw • pertamina/gld/uplink
- **Proses**
decrypt • integrity • replay • topology
- **Output**
decoded • alarm • status • events • integrity

MQTT
NORMALIZED →

gasleakdetectionV2-April

- **Input wajib**
adapter mengubah MQTT normalized → addReading/device
- **Layanan**
NestJS GraphQL • Prisma • database RU
- **Konsumen**
Next.js dashboard • map • alerts • analytics

MANDATORY DATA CONTRACT

deviceId • timestamp • gasClass/aiClass • confidence • battery • network/health • messageId untuk deduplication.

Frontend — RU operator UI dan HQ read-only UI

IMPLEMENTED: Next.js client dengan GraphQL polling. GraphQL subscription tidak diimplementasikan.

FRAMEWORK

Next.js 15 • React 18 • TypeScript

VISUALISASI

Mapbox GL • Recharts • Tailwind CSS

DATA ACCESS

GraphQL Request → endpoint RU :4000/graphql

REFRESH

Polling berkala; default setting 10 detik

MODUL

Overview • Devices • Map • Alerts • Events

ANALITIK

Timeline sensor • confidence • status perangkat

MODE RU

Operasional lokal termasuk menu admin/settings

MODE HQ

Read-only; mutation dan settings disembunyikan

KILANG-SUPPLIED ACCESS

Kilang menetapkan internal URL, DNS/IP, TLS certificate, identity/access policy, operator workstation, dan availability target.

Backend: ingestion, storage, dan API GraphQL

IMPLEMENTED RU: SQLite. PostgreSQL/TimescaleDB belum menjadi production baseline yang disetujui.

RUNTIME/API

Node.js • NestJS • Apollo GraphQL • port 4000

DATA LAYER

Prisma ORM • current RU schema: SQLite file-based

CORE ENTITIES

Device • GasReading • SystemSettings • EventLog • User

INGESTION

addReading validates confidence 0–1 and derives LOW/MIDDLE/HIGH

ALERT LOGIC

Threshold default warning 0.70 • critical 0.80 • dedup 15 menit

API

dashboard • devices • timeline • events • analytics • gatewayPull

SECURITY STATE

Production hides playground/introspection; CORS still broad

OPEN ITEM

MQTT adapter, backup/retention RU, auth/TLS and hardening

DATABASE BASELINE — OPEN

Tetapkan SQLite untuk pilot atau migrasi PostgreSQL. Keputusan wajib mencakup sizing, backup, retention, HA, dan recovery objective.

DFD-02: RU → VPS/HQ (NOT COMMISSIONED)

RU beroperasi mandiri; VPS menarik data read-only melalui VPN. Remote actuation tidak termasuk scope fase ini.



VPS RETENTION TARGET

Reading dan non-critical event: 7 hari. Critical event: 90 hari. Final retention policy memerlukan approval Kilang.

NOT COMMISSIONED

Gateway service tersedia di repo. Production deployment, VPN, credential, DNS, backup, dan monitoring belum diverifikasi.

GLD Node: hardware, firmware, dan telemetri

Firmware v0.8.14 • 8-channel MQ acquisition • on-device inference • 24 VDC $\geq$ 2 A.

MCU

ESP32-S3-WROOM-1U-N16
R8

ADC

ADS1256, 24-bit

SENSOR

8 × MQ

TELEMETRI

LoRa tiap 10 detik

ALARM

gasClass $\neq$ 0; confidence
 $\geq$ 30%

KEAMANAN FRAME

AES-128-GCM, 39 byte

AI

CNN Dual-Branch, int8

CATU DAYA

24VDC $\geq$ 2A per unit

MOUNTING

L-bracket referensi Emerson

* 99,20% berasal dari laporan model int8. On-device execution memerlukan traceability ke firmware commit dan binary release.

Cluster Head dan Gateway — functional specification

Cluster Head menyediakan cache/relay LoRa; Gateway menyediakan LoRa-to-MQTT bridge. Seluruh buffer bersifat volatile.

CLUSTER HEAD

- **Dual SX1262**
STAR ke GLD + MESH ke CH/Gateway
- **Cache 32 node**
Failover parent otomatis berbasis RSSI
- **2 panel solar + 18650**
Surplus daya 1,5–4,4 Wh/hari
- **16 unit tersedia**
9 CH besar + 7 CH kecil

GATEWAY

- **Radio mesh 921,0 MHz**
SF9 • 17 dBm
- **Bridge Wi-Fi/LAN ke MQTT**
MAC didaftarkan; satu subnet dengan server
- **Buffer 8 item RAM**
Volatil, bukan penyimpanan permanen
- **Unit indoor**
Antena ditempatkan outdoor/rooftop

CAPACITY LIMIT — OPEN

Maximum supported GLD Node per Cluster Head dan per Gateway belum ditetapkan melalui load test.

STAR dan MESH menggunakan domain radio terpisah

Frekuensi, spreading factor, dan sync word dikonfigurasi berbeda untuk mengisolasi kedua link.

PARAMETER	STAR — GLD ↔ CH	MESH — CH ↔ GW
Frekuensi	920,0 MHz	921,0 MHz
Spreading Factor	SF7	SF9
TX power	17 dBm	17 dBm
Sync word	0x12	0x34

100 m terbuka		PDR 100%
242 m Gerbang Utara		PDR 100%
200 m terhalang gedung		PDR 87%

Instalasi permanen dilarang sebelum sertifikasi

Enclosure menggunakan dua kompartemen dengan seal/cable gland; compliance ATEX belum diverifikasi.

INSTALLATION HOLD

ATEX certificate dan elevated-mounting approval belum tersedia. Permanent installation tidak diizinkan.

MODUL ELEKTRONIK

PCB + power di enclosure Ex-Proof

SEAL / CABLE GLAND →

SENSOR CARTRIDGE

Elektronik sensor di kompartemen terpisah

- **Non-intrusive mounting**
L/U-bracket pada existing structure; drilling dan welding dilarang
- **Hazardous-area power protection**
24 VDC protection wajib sesuai zone; Gateway ditempatkan di safe area

Model AI tidak mencakup tiga gas requirement RU

Current model classes: LPG, CO₂, H₂, dan clean air. Benzena, CO, dan H₂S tidak tercakup.

AKURASI MODEL INT8

99,20%

UJI REAL-TIME

97,65%

UKURAN MODEL

9,14 KB

KELAS MODEL SAAT INI

LPG

CO₂

H₂

Udara Bersih

REQUIREMENT BELUM DICAKUP

Benzena

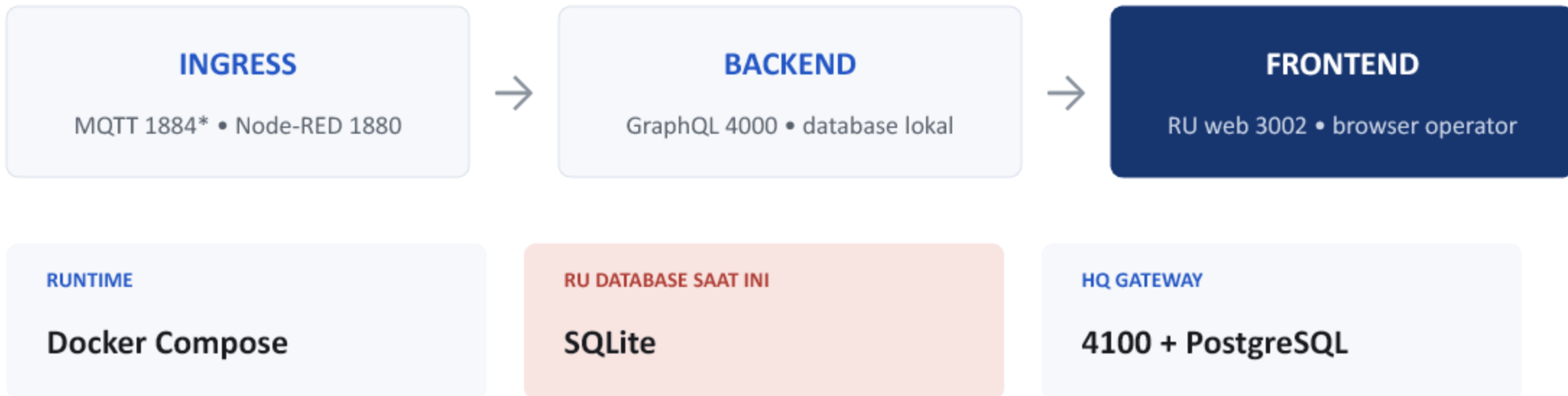
CO

H₂S

Metana belum
tervalidasi

RU services pada server/VM internal Kilang

Port yang ditampilkan adalah current repo/bench defaults. Production firewall rules ditetapkan IT/OT Kilang.



MINIMUM FIREWALL POLICY

Allow: Gateway→MQTT, operator→frontend, frontend→backend, dan RU→VPS via VPN jika disetujui. Deny public access ke Node-RED admin.

Baseline 24 VDC; portable battery tidak digunakan

Cluster Head menggunakan dual solar panel dan 18650 battery; hazardous-area certification belum tersedia.

GLD NODE

- **Input 24 VDC ≥ 2 A per unit**
Production power baseline
- **Cable dan PSU disediakan RU**
Satu protected circuit per installation point
- **Portable battery excluded**
Measured endurance 1,76 hari; requirement 30 hari

CLUSTER HEAD

- **Dual-panel energy surplus**
Calculated surplus 1,5–4,4 Wh/hari
- **Panel → BQ25185 → 18650**
TPS63020 3,3V menuju beban
- **Solar/battery certification unavailable**
Hazardous-area verification mandatory



Baterai GLD: 1,76 / 30 hari

Delapan open item memblokir commissioning

Tidak ada open item pada slide ini yang dapat ditutup hanya dengan hasil uji laboratorium.

- **TRA/JSA approval**

OPEN — internal draft only

- **ATEX certification**

OPEN — permanent installation prohibited

- **MQTT ingestion adapter**

NOT IMPLEMENTED

- **Gateway/server network approval**

OPEN — VLAN, ACL, identity, firewall

- **Elevated-mounting approval**

OPEN — test limited to human height

- **Gateway/CH capacity limit**

OPEN — load test not available

- **RU production database**

OPEN — SQLite pilot vs PostgreSQL

- **RU server/VM allocation**

OPEN — resource confirmation required

PROGRESS BOUNDARY

≈39% merepresentasikan engineering laboratory progress; tidak merepresentasikan certification, commissioning, atau operational acceptance.

Minimum commissioning acceptance criteria

Sistem hanya dapat dinyatakan operational setelah seluruh evidence diverifikasi di RU.

1

SITE READINESS

TRA/JSA approved; mounting structure released; protected 24 VDC circuit available.

2

SYSTEM INTEGRATION

Server/VM allocated; network path approved; MQTT adapter and RU database operational.

3

SAFETY COMPLIANCE

ATEX approval issued; required gas classes validated; permanent installation authorized.

MANDATORY EVIDENCE

MQTT broker online; ingestion tervalidasi; alarm tampil di dashboard; restore test lulus; hazardous-area approval diterbitkan.